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ON STEADY FLOW.

By Lowen, of White Tower University.

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The following paper was read:—

ON STEADY FLOW.

By Lowen, of White Tower University.

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SUMMARY. A mana pivot that adjusts its intake against the deviation of its own output, in proportion to that deviation, may hold that output steady in the tides; yet on the trial benches some such returns steady the output, and others make it swing ever wider. Without entering into the carving of the return-pattern, this paper treats the dynamics of the return. While the deviations are small and the return does not exceed the capacity of the pivot, the motion of the deviation may be resolved into parts, each of which either increases continually, or diminishes continually, or swings with increasing amplitude, or swings with diminishing amplitude; the condition of steadiness is that every part be of the second or the fourth kind. When the pivot's own response, the delay of the reading and the delay in changing the intake are taken together, the deviation satisfies an equation of the third degree, and the condition of steadiness is that the product of the two middle coefficients exceed the product of the two extreme ones; when the three delays are equal, the strength of the return must be less than eight. The stronger the return, the smaller the deviation the tide leaves behind, but the nearer the bound. To remove even this deviation an accumulating part must be added; if it accumulates too fast, the output again swings ever wider, and its bound is given. These reckonings agree in the main with the records of six trial benches at Tallin, Kanaich and White Tower. At the end a method is given by which craftsmen may measure the delays and fix the strength, and the places the reckoning does not reach are stated.

1. The question

Severin's Fourteenth Question asks: "Can a mana pivot, under the shock of the tides, hold its own output steady?" [1] For seven hundred years every city has answered it with men: when the tide comes, craftsmen stand by the pivot, watching the output and adjusting the intake. In the year of the Great Tide the engines of several cities were very nearly not held; I was then at White Tower, and saw craftsmen who did not leave the pivot for three days and nights. The year F.E. 24 was the third tide year since the Great Tide, and in its Month of Scrolls the city engine of Tallin twice ran now strong and now weak upon the crest of the tide; the craftsmen on watch took two days to bring it level [9].

The thought of letting the pivot change its own intake according to the deviation of its output is not new. Since the year 24 the craftsmen of Tallin and Kanaich have tried several carvings upon trial benches, and the workshop of White Tower University has made two benches for me [10]. The results are perplexing. Some pivots, once given a

return, are left with only a small deviation of output by the push of the tide; others swing ever wider, so that the return must be cut off; and the same carving, cut a little stronger, passes from the first sort to the second. The craftsmen say that the return must be "just right", but no one can say how much that is.

I propose, without entering into the manner in which a return-pattern is carved, to treat only of how the pivot moves when it has one. This is the heart of Severin's question: whether a pivot can hold itself steady depends not upon whether a return can be carved, but upon whether, once carved, it steadies.

Canning reckoned how the lift is shared out anew among the groups of arrays when a city leans [3]. He asked where the city comes to rest, not how it moves before it rests; and the second is what is asked here. That a machine can rest at a certain place, and that it will return to that place when disturbed, are two different things.

2. The pivot and its return

Let the output of the pivot exceed its set value by y, which I call the deviation of output, and let the intake exceed its set value by u. When the intake changes, the output does not follow at once but arrives after a little while; and the tide also pushes the output. Let u be measured in such units that a change u of intake changes the output in the end by u, and let the push of the tide be w, that is, the change the tide would make in the output if the intake were held fixed. The tide for the most part changes slowly: the ten-year tide, the several constituent tides and the rise and fall within a day are all far slower than the delays of a pivot [4]; only when the head of a tide comes suddenly does it act like a push suddenly applied. The pivot's own response is written

$$\tau_0 \frac{dy}{dt} + y = u + w \tag{1}$$

and τ_0 is called the pivot's own time of response. The return-pattern reads the output, and the reading too cannot follow at once, but follows by degrees according to (2). Let the reading be z:

$$\tau_1 \frac{dz}{dt} + z = y \tag{2}$$

where τ_1 is the delay of the reading. By a delay I mean throughout such a time of response, in which a quantity follows by degrees, and not a waiting after which it leaps at once. When the deviation has been read, the intake is changed, and this too cannot be done in an instant:

$$\tau_2 \frac{du}{dt} + u = -Kz \tag{3}$$

where τ_2 is the delay in changing the intake, and K the strength of the return: for each part added to the reading, the intake is in the end diminished by K parts. The pivot, the reading-pattern and the intake-pattern need not follow in exactly this manner; I write them so because it is the simplest manner, because it can be measured on the bench, and because the following measured there agrees with it (Section 7). The three

are supposed besides to act independently and to remain unchanged during a trial, and the tide to enter only as a push added to the output.

These three equations suppose two things more. First, that the deviations are small, so that everything is proportional to them. Secondly, that the intake the return calls for lies, at every moment of the passing motion, within the capacity of the pivot. By capacity I mean here the upper and lower bounds of what the pivot can take in, in a unit of time: u may be positive or negative, and a greater intake meets the upper bound, a lesser the lower. The capacity depends upon the fuel and upon what is left in the cores, and is not the same thing as core capacity [5], which is the whole energy a core delivers when it is spent; in the trials the cores of every bench were more than half full. If the tide is too violent, and the return calls for an intake beyond these bounds, the intake stands at the bound and no longer follows the reading; equation (3) then fails. Everything reckoned below holds within these two conditions only.

The return-pattern reads the output by a sensing piece such as that of a mana meter [2], set however upon the conduit of the output, so that it reads the mana delivered to the load and not the free mana round about. On the bench it is first set true against known loads, and a second piece, not joined to the conduit, is set beside it, so that the direct effect of the mana round about may be taken out of the reading. The reading must further be compared with a standard. Taunton has shown that standards disturbed by the tide are lastingly changed, in capacity and in response; for the set he compared, the ratio of after to before was 1.011 [8]. The "steadiness" of this paper is therefore steadiness with respect to the standard: if the standard changes, the output changes with it; and a sensing piece that has passed through a tide may have changed its time of response, and must be measured again.

3. The four kinds of motion

Let w be constant. When the fixed deviation it leaves has been taken away, the solution of equations (1) to (3) is a sum of terms, each of the form e^{nt}, where n is a root of a certain equation (where a root is repeated, the term is multiplied by t, t^2 and so on, but so long as the real part of the root is negative such terms die away as well). The roots may be real, or they may be complex in pairs. The motion of the deviation may therefore be resolved into parts, the amplitude of each being of one of these four kinds:

First, it may increase continually; secondly, it may diminish continually; thirdly, it may swing to and fro with increasing amplitude; fourthly, it may swing to and fro with diminishing amplitude. At the boundary there is a fifth: it may swing to and fro, neither growing nor dying (Section 5).

The first and third kinds are inconsistent with steady flow; a good return admits only the second and fourth. In the language of reckoning: every real root must be negative, and every complex root must have its real part negative.

4. When there are only one or two delays

If there is no return, $K = 0$, the deviation tends to w in a time of the order of τ_0: as far as the tide pushes, so far the output departs.

If the intake can be changed without any delay, $\tau_2 = 0$, and n satisfies $(\tau_0 n + 1)(\tau_1 n + 1) + K = 0$. The sum of its two roots is $-\frac{\tau_0+\tau_1}{\tau_0\tau_1}$, which is negative, and their product is $\frac{1+K}{\tau_0\tau_1}$, which is positive; so the two roots are either both negative and real or a pair of complex roots with negative real part. However strong the return, the output will not swing ever wider. Once the two roots are complex, their real part is always $-\frac{\tau_0+\tau_1}{2\tau_0\tau_1}$: a stronger return makes the swinging closer, but the rate at which the swinging dies away is unchanged.

Yet the pivots on the benches do swing ever wider. The reason is that the intake cannot be changed in an instant, and the reading too takes time to follow; the delay in changing the intake is never nothing. The returns treated in this paper have at least three delays, and to the case of three I now turn.

5. Three delays

Putting e^{nt} in equations (1) to (3), we find that n must satisfy

$$(\tau_0 n + 1)(\tau_1 n + 1)(\tau_2 n + 1) + K = 0 \tag{4}$$

which, when expanded, is

$$an^3 + bn^2 + cn + d = 0 \tag{5}$$

where $a = \tau_0\tau_1\tau_2$, $b = \tau_0\tau_1 + \tau_0\tau_2 + \tau_1\tau_2$, $c = \tau_0 + \tau_1 + \tau_2$ and $d = 1 + K$; all four coefficients are positive.

The question is how great K must be before (5) has a root with positive real part. Let K increase from nothing. When $K = 0$ the three roots are $-\frac{1}{\tau_0}$, $-\frac{1}{\tau_1}$ and $-\frac{1}{\tau_2}$, all negative. As K increases the roots move continuously; a real root cannot pass through zero, because at $n = 0$ the left side of (5) is d, which is positive. If a root is to acquire a positive real part, therefore, a pair of complex roots must cross the imaginary axis. At the moment of crossing n is purely imaginary, $n = \omega i$, and the deviation neither grows nor dies but swings to and fro with the period $2\frac{\pi}{\omega}$. Putting $n = \omega i$ in (5), the real and the imaginary parts must vanish separately:

$$b\omega^2 = d, \quad a\omega^2 = c$$

whence, dividing the one by the other, $bc = ad$ at the boundary; (5) then falls exactly into the factors $(an + b)\left(n^2 + \frac{c}{a}\right) = 0$, which give, besides a negative real root, the pair of purely imaginary roots $\pm i\sqrt{\frac{c}{a}}$. Below the boundary the real parts of all three roots are negative; and as the boundary is passed, the real part of this pair increases with K, for at $n = \omega i$ the real part of the rate at which n changes with K is $\frac{c}{2(c^2+b^2\omega^2)}$, which is positive. Thus:

The condition of steadiness. The product of the two middle coefficients must exceed the product of the two extreme ones,

$$bc > ad \tag{6}$$

that is, the strength of the return must be less than

$$K^* = \frac{bc}{a} - 1 = \frac{(\tau_0\tau_1 + \tau_0\tau_2 + \tau_1\tau_2)(\tau_0 + \tau_1 + \tau_2)}{\tau_0\tau_1\tau_2} - 1 \tag{7}$$

When the three delays are equal, $K^* = 8$ (the delays are here supposed greater than nothing, and K not less than nothing). When one delay is far less than the other two, K^* is very great, which is the case of Section 4. The more unequal the delays, the wider the bound; the more nearly equal, the narrower.

The good of the return is that the deviation the tide leaves behind is made smaller. While w is constant, and (6) holds, the deviation comes at length to rest at

$$y = \frac{w}{1 + K} \tag{8}$$

The stronger the return, the smaller the deviation left; but the stronger it is, the nearer the bound; near the bound the swinging dies away slowly, and beyond it the output swings ever wider. Between these two lies what the craftsmen call "just right".

6. The period at the boundary, and how to measure the delays

Upon the boundary the deviation swings to and fro, neither growing nor dying, with the period

$$P = 2\pi\sqrt{\frac{a}{c}} \tag{9}$$

Near the boundary the period of the swinging is not far from this; farther from it, both the period and how much of each swing remains after the next must be found from the complex roots of (5), and where there are no complex roots there is no swinging.

A craftsman who would use this condition must first measure the three delays. It may be done thus, the return being first cut off and the other parts held fast, so that each part is measured by itself. Change the intake suddenly by a little, and note how long the output takes to cover sixty-three parts in a hundred of its whole change, that is, one less the reciprocal of e: that is τ_0. Let the output change suddenly by a little, and note how long the reading takes to cover the same part: that is τ_1. Hold the reading, change it suddenly by a little, and note how long the intake takes to cover the same part: that is τ_2. The strength K is measured besides at rest: hold the reading, change it by a little, and note by how much the intake changes in the end. Reckon K^* by (7), and make the strength of the return about one half of it; there is then room to spare.

A craftsman standing by a pivot is himself a kind of return. He sees the output, considers, and then changes the intake with his hand, and each of these three steps has its delay. The old craftsmen say "better slow than violent": a violent craftsman adjusts too far, and the supply comes now strong and now weak, which is just what Severin described in his Fourteenth Question. The condition of this paper agrees with the old saying: whether the return is a man's hand or a return-pattern, if its strength passes the bound the output swings ever wider. Why a man's hand should be harder to keep within the bound than a pattern, I cannot say, for this paper has written no equation for a man's hand.

7. Comparison with the trial benches

Table I gives the records of six trial benches. The two of Tallin were made in the workshop of Arundel, and are one pivot, the return-pattern of the second being cut

deeper; the two of Kanaich are two pivots made by craftsmen of Kanaich; the two of White Tower were made by the workshop of White Tower University for this paper, and are again one pivot with the strength of the return changed [10]. The three delays were measured by the method of the last section, the error of measurement being within about a tenth at each bench; K was measured at rest by the method of the last section, and differed from what the carving gives by less than one part in twenty. At every bench the push applied was two parts in a hundred of the set output and the load six tenths of the full load, and each trial was made three times; the table gives the middle of the three. The reckoned columns are found from (7) and from the roots of (5). The swinging is counted as at rest when its amplitude has fallen below a twentieth of its first.

The ratio per swing is the ratio of the amplitude of one swing to that of the swing before: below one the swinging dies away, above one it grows.

TABLE I. The delays, return and motion of six trial benches (times in seconds; the ratio per swing and the period are reckoned)

Bench	τ_0	τ_1	τ_2	K	K^*	*Ratio per swing*	*Period*	*Observed*	*Observed period*
Tallin A	6	2	3	4	10	0.27	15.7	steady	about 16
Tallin B	6	2	3	12	10	1.24	10.7	wider	about 11
Kanaich A	8	1	4	9	16.9	0.45	12.7	steady	about 13
Kanaich B	8	4	4	9	9.0	1.00	17.8	slight growth	about 18
White Tower A	5	5	5	6	8	0.69	20.0	steady	about 21
White Tower B	5	5	5	10	8	1.30	16.8	wider	about 17

At Tallin B the return was cut off at the fifth swing; White Tower A came to rest after eight or nine swings.

At the five benches that can be judged, the steadiness or unsteadiness agrees with the reckoning, and the periods differ by less than a tenth. The bench Kanaich B lay upon the boundary, and its amplitude grew only a little in a quarter of an hour; the error of the delays is enough to put it on either side of the boundary, and this bench cannot be judged. The records suffice to show whether the ratio per swing is greater or less than one, but the ratios in the table are reckoned, and cannot yet be checked by measure. The two benches of Tallin are one pivot; only the return of the second is cut deeper, and its K passes the bound, so that it passes from steady to unsteady, which is precisely what perplexed the craftsmen; and so too with the two of White Tower.

Fig. 1 is reckoned step by step from (1) to (3) and from (10) below: the three delays are each five seconds, all deviations are nothing at first, at the fifth second the push of the tide is suddenly increased by one part, and y is measured in that part. With $K = 4$ the output swings a few times and comes to rest at one fifth; with $K = 8$ it lies upon the boundary and swings without ceasing; with $K = 10$ it swings ever wider, and its

line is drawn only while the deviation is still small, beyond which the supposition of small deviations fails.

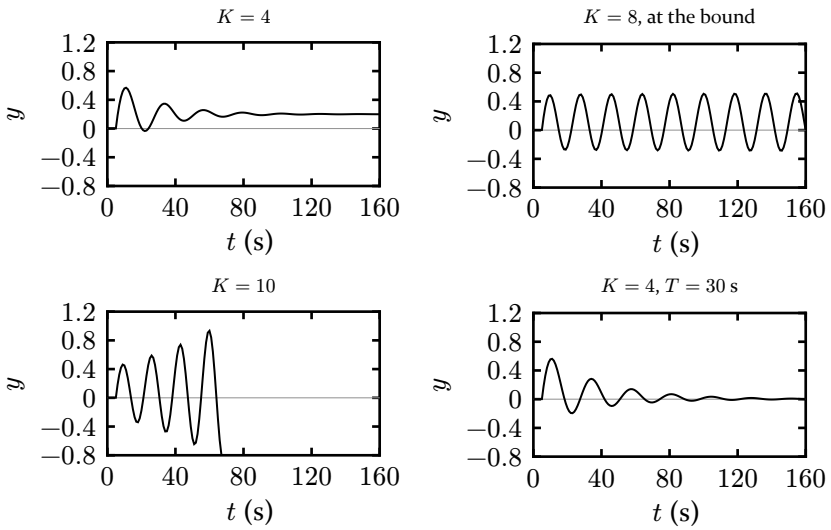


Fig. 1. The deviation of output, when the three delays are each five seconds and the push of the tide is suddenly increased by one part

8. The accumulating part

By (8), a return always leaves a deviation $\frac{w}{1+K}$. To remove this too, a further part may be added to the return: so long as a deviation remains, it is accumulated, and the intake is changed further according to what has been accumulated. I call this the accumulating part. Equation (3) becomes

$$\tau_2 \frac{du}{dt} + u = -Kz - \frac{K}{T}\int z dt \tag{10}$$

where T is the time of accumulation; the smaller T, the faster the accumulation. For a constant push, so long as (12) below holds and the intake the compensation calls for lies always within the capacity, the deviation now comes at length to nothing; if the tide changes slowly, the deviation only follows the change, and cannot be said to be wholly removed. The number n must satisfy

$$an^4 + bn^3 + cn^2 + (1+K)n + \frac{K}{T} = 0 \tag{11}$$

Seeking the boundary as in Section 5, by putting $n = \omega i$, we find from the imaginary part $\omega^2 = \frac{1+K}{b}$, and from the real part $a\omega^4 - c\omega^2 + \frac{K}{T} = 0$. When T is very great, the added root is about $-\frac{K}{T(1+K)}$ and the other three are the roots of (5); so long as (6) holds, then, the motion is steady when T is great, and as T is made smaller the first crossing of the imaginary axis falls exactly where the following becomes an equality. The condition of steadiness, K and T being greater than nothing, thus consists of two parts: first, that (6) still hold; secondly, that

$$(bc - a(1+K))(1+K) > b^2 \frac{K}{T} \tag{12}$$

that is, the accumulation may not be too fast, and T must exceed $b^2 \frac{K}{(bc-a(1+K))(1+K)}$. When the three delays are each five seconds and $K = 4$, T must exceed nine seconds. If the accumulation is too fast, it presses past before the deviation has come back, and the output swings ever wider; if it is slow, the deviation is slow to come back to nothing. The fourth line of Fig. 1 is the case $K = 4$, $T = 30$ seconds, in which the deviation comes at length to nothing. The accumulating part has not yet been tried upon a bench; what is said of it here is reckoning only.

9. Two regulations together

When Oldham made trial of the joint tuning of the mana net in a quarter of Kanaich, the regulation of the arrays and the regulation of the engines for a time worked against each other, and the arrays swung to and fro; it was afterwards arranged in layers, the arrays first levelling themselves where they stood and the engines then adjusting the whole output slowly, and so it became steady [7]. Two regulations acting upon the same thing may each lie within its own bound, and the two together pass beyond it. The simplest case is this: if two returns read the same output and move alike, together they are one return of strength $K_1 + K_2$; K_1 and K_2 may each be less than K^*, and their sum greater. Where the speeds differ, or they act upon different places, the equations of their joining must be written out, and this paper does not treat them. When the regulations were arranged in layers, the arrays came to rest first and the engines were then adjusted slowly; the reason is akin to that of Section 8, that the accumulation must be slow, but the joining is different, and (12) cannot be carried over to it: why Oldham's layers are steady remains to be reckoned. So also with parallel pivots. Brandon carved a pattern of even flow between parallel pivots, so that they share the load in proportion [6]. If the pivots are alike and move alike, and the returns of all read the same whole output, the strength of them together is the sum of their strengths, and must be fixed for that sum. The general case wants its own equations; and that the whole output does not swing does not mean that no pivot swings, for the pivots may swing one against another, one rising as another falls, and it is just this that the pattern of even flow governs.

10. What the reckoning does not reach

This paper asks only after the motion when the deviations are small and the return does not exceed the capacity of the pivot. If the tide is too violent and the return calls for more intake than the capacity, the intake stands at its capacity, and the equations above no longer hold. While it so stands the return is in effect weakened, but the accumulating part goes on accumulating, and when the intake leaves the bound what has been accumulated may be too much. I guess that it is then as if the delays were lengthened; but this is a guess only, and what then follows I cannot yet reckon.

For the equation of the third degree the condition of steadiness can be judged from the coefficients alone, namely (6); for the equation (11) of the fourth degree it can be found by the method of Section 5. For equations of higher degree the boundary may

be found in the same way, but each equation must be taken by itself; a general method by which one may judge from the coefficients alone whether every root has its real part negative I have not found. When several regulations and several pivots are joined together, the degree of the equation soon grows high. This I leave to the reckoners.

The return-pattern is carved at the intake of the pivot; how it is carved this paper does not treat. As things now stand, it must be carved anew for each kind of engine.

Acknowledgments. Arundel and several craftsmen of Kanaich lent me the records of their trial benches, and measured the delays again at my asking; the two benches of the workshop of White Tower University were kept by the apprentices of the workshop in turns.

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